JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/03

Applications Paper and Planning Question **14 September 2015**

Paper 3 **2 hours**

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do no use staples, paper clips, glue or correction fluid.

Answer **all** questions.
The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
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Total	

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Answer **all** the questions in this section.

1 Human insulin can be synthesised in a laboratory strain of *E. coli* using recombinant DNA (rDNA) technology. The starting point for the process is mature insulin mRNA, isolated from human pancreatic cells.

(a) Explain what is meant by *recombinant DNA (rDNA)*. [2]

1. Recombinant DNA refers to the joining / annealing of DNA;;
2. From more than one source eg, prokaryotic DNA and eukaryotic DNA / plasmid and insulin gene or GOI;;

(b) State the role of each of these enzymes in producing rDNA carrying the insulin gene. [2]

(i) reverse transcriptase

1. Catalyse formation of phosphodiester bonds between adjacent deoxyribonucleotides/DNA nucleotides resulting in synthesis of cDNA strand (complementary to mRNA);;

(ii) restriction enzyme

2. Cut plasmid <u>and</u> GOI by hydrolyzing / breaking phosphodiester bonds between adjacent nucleotides in DNA;;

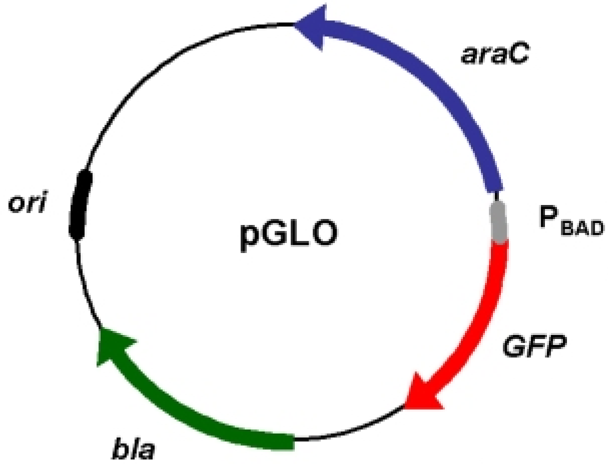
(c) A DNA library is a collection of cloned DNA fragments. The two types of DNA libraries, genomic library and cDNA library, allows one to isolate a specific gene for use in rDNA technology.

Explain why insulin gene is isolated from a cDNA library instead of a genomic DNA library. [2]

1. cDNA library would have many copies (of the coding sequence of) insulin gene while genomic DNA library would contain one copy of the insulin gene;;
2. insulin genes are intact in a cDNA library while insulin gene obtained from genomic DNA may not be a single functional fragment (may be cut internally more than one time by the restriction enzyme used to cleave the genomic DNA) / not intact;;
3. cDNA library contains only coding regions of insulin gene (as introns have been removed) while genomic insulin gene may contain introns;;
4. Smaller amount of DNA in a cDNA library and thus easier to isolate insulin gene while the genomic library may contain too many DNA fragments resulting in a more tedious screening process to isolate insulin gene;;

Plasmid vectors such as pGLO are used to introduce insulin genes into *E. coli* by transformation and enable replication of these insulin genes.

In the synthesis of pGLO, the genes that code for arabinose catabolism (*araB*, *araA*, and *araD*) have been replaced by the green fluorescent protein (GFP) gene. Colonies appear fluorescent green under UV light when GFP gene is expressed and white when GFP gene is not expressed. The multiple cloning sites is found within the GFP gene. In the presence of arabinose, the arabinose interacts directly with AraC protein. AraC protein change shape and promotes the binding of RNA polymerase to P_{BAD}.



araC - gene for araC regulatory protein

GFP - gene for green fluorescent protein under control of P_{BAD}

bla (*β-lactamase*) - gene for ampicillin resistance

P_{BAD} - arabinose promoter

Fig. 1.1

(d) With reference to pGLO shown in Fig. 1.1, describe two properties of plasmids that allow them to be used as cloning vectors. [4]

1. **pGLO contains an origin of replication;;**
2. **so that the vector and the inserted gene of interest / insulin gene can replicate independently of the bacteria chromosome to produce multiple copies within the host cell;;**

3. **pGLO contain genetic / selectable markers;;**
4. **e.g. *β-lactamase / bla* gene that confer resistance of the host cell to ampicillin OR green fluorescent protein / GFP gene coding for green fluorescent protein for which enable selection;;**
5. **Allows for the selection of successfully transformed bacteria containing both recombinant and non-recombinant plasmid;;**

(1 e.g. of markers)

6. **Plasmids have multiple cloning sites (MCS) that contains many restriction enzyme sites;;**
7. **which can be recognized, bound and cut by many different restriction enzymes for insertion of insulin gene;;**

(e) After the transformation process, bacterial suspensions are spread onto plates X and Y.

(i) Complete Table 1.1 to show the observations made of each plate when exposed to UV light. [2]

Table 1.1

plate	content	observation(s)
X	nutrient rich media / Amp	**White colonies;;**
Y	nutrient rich media / Amp / arabinose	**Fluorescent, green colonies;;** **White colonies;;**

(ii) With reference to your observations in Table 1.1, explain which plate(s) may contain colonies that have taken up the insulin gene. [3]

1. **Both plates X and Y;;**
2. **(Transformed) bacterial cells that have taken up the insulin gene / containing recombinant plasmids appear as white colonies;;**
3. **This is because insulin gene is inserted within *GFP* gene / *GFP* gene disrupted, insertional inactivation of *GFP* gene occurs and no / non-functional GFP proteins are produced;;**
4. **Arabinose is not present in plate X and GFP gene will not be expressed whether transformed bacteria with recombinant and non-recombinant plasmids are present;;**

[Total: 15]

2 Cystic fibrosis (CF) is caused by mutations within the gene encoding the cystic fibrosis transmembrane conductance regulator (CFTR) protein. The most common mutation in this gene results in the loss of one amino acid from the CFTR protein. This deletion, ΔF508, accounts for about 70% of mutations in cystic fibrosis.

(a) (i) State the type of mutation responsible for this change in the amino acid sequence. [1]

1. **deletion of 3 nucleotides CTT;;**

(ii) Outline the role of the product of the normal *CFTR* allele. [3]

1. **Normal CFTR protein is a transmembrane chloride channel;;**
2. **that allows the efflux / transport of chloride ions out of the epithelial cells;;**
3. **The flow of chloride ions helps control the movement of water in tissues/cells of airways / lungs / digestive system / pancreas;;**
4. **water moves out of cell (down water potential gradient by osmosis), watery mucus produced/thinning the mucus;;**
5. **(regulates the fluid consistency of mucus).**

Since the *CFTR* gene was discovered and cloned, gene therapy to introduce a normal copy of the gene has been a possibility for CF treatment.

In 2012, the UK Cystic Fibrosis Gene Therapy Consortium launched a trial involving 140 patients to assess the clinical efficacy of a non-viral CFTR gene-liposome complex as a delivery system for a functional CFTR gene. 78 of whom received at least 9 monthly doses of the therapy, while a control group of 62 inhaled saline solution. The results from the year-long trial, showed that those who had been breathing in the gene treatment had, on average, a 3.7 percent improvement in their lung function compared with the placebo group.

Over the course of the year, the lung function of patients in the treatment group seemed to stabilise, while the control group showed a steady decline in lung power.

(b) (i) Outline how CFTR gene-liposome complex can be used as part of a gene therapy treatment for CF. [3]

1. **The CFTR gene-liposome complex is sprayed into the nose/mouth of CF patients in the form of <u>an aerosol nasal spray to target cells/epithelial cells of respiratory tract/lungs</u>;;**
2. **Liposomes enter the epithelial cells via endocytosis / lipofection / fuse with the cell membrane of epithelial cells;;**
3. **The <u>normal functional CFTR allele is released into the cytoplasm of cells</u> and subsequently enters the nucleus and is expressed to produce normal functional CFTR protein;;**
4. **The CFTR protein embeds into cell membrane and begins to transport chloride ions out of cells, thinning the mucus (OWTTE part (b), pt 4) (water moves out of the cell and dilute the abnormally thick sticky mucus and alleviating CF symptoms);;**

(ii) Account for the 3.7 percent improvement in the patients' lung function. [1]

(Why there is improvement)

1. **Some epithelial cells had taken up / contains the normal functional CFTR allele during the non-viral / CFTR gene-liposome complex gene delivery;;**

(Why insignificant improvement)

2. **Most normal functional allele did not end up in the nucleus / end up in the cytosol and does not integrate into target cell genome / difficulty integrating normal functional allele into target cell genome;;**
3. **Problems with controlling the activity of gene expression;;**
4. **Normal functional allele is degraded by nucleases and does not integrate into target cell genome;;**
5. **Majority of epithelial cells are still lacking the normal functional transmembrane protein;;**

In 2005, a group of scientist explored using adult stem cells from bone marrow, referred to as marrow stromal stem cells (MSCs), to provide a therapy for Cystic Fibrosis (CF). They found that MSCs can be induced to differentiate into airway epithelial cells.

The MSCs are amenable to *CFTR* gene correction using adenoviral vector. This serves as another attractive potential method of correcting the basic defect in respiratory epithelium.

(c) (i) Describe the features of MSCs and their role in CF gene therapy. [3]

1. **MSCs are multipotent / exhibit multipotency (marrow stromal stem cells) have ability to differentiate into a limited range of cell type and so are not pluripotent or totipotent;;**
2. **Undifferentiated cell and can remain unspecialized for a long time;;**
3. **Capable of dividing indefinitely and renewing itself;;**

(pt 1-3: max 2)

Role

4. **Can undergo differentiation giving rise to airway epithelial cells;;**
5. **producing copies of itself with the normal functional allele;;**

(ii) Explain the advantages of using an adenoviral vector as the gene delivery system for the normal allele of the *CFTR* gene. [2]

1. **Specific targeting, the adenoviral vector is able to deliver genes to specific target cells / tissues - epithelial cells / epithelial tissue / cells of the respiratory tract (any one);;**
2. **Greater efficiency, the adenoviral vector is able to transfer the gene into the target cell more effectively;;**
3. **able to introduce/deliver the normal allele of the *CFTR* gene into the nucleus;;**

[Total: 13]

3 Bananas are the second largest food-fruit crops of the world produced in the tropical and subtropical regions of mostly the developing countries. Bananas are important as a crop but the planting material required is often in short supply. Scientists have developed a micropropagation system as shown in Fig. 3.1.

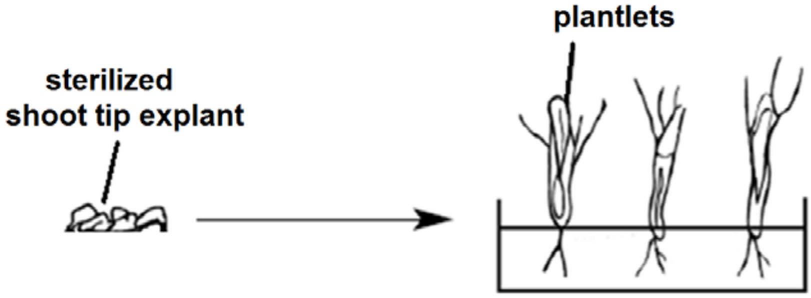


Fig. 3.1

(a) With reference to Fig. 3.1, outline the techniques of cloning plants from tissue culture. [3]

1. **The shoot tip explant is then aseptically transferred to culture vessels containing aseptic culture medium that contains nutrients (e.g. K^+, sucrose) and plant growth regulators (auxin and cytokinin);;**
2. **Auxin and cytokinin in equal ratio is added to the nutrient agar to stimulate the cells of the explant to divide by mitosis to form a callus;;**
3. **As callus increases in size, it can be subcultured into many calli by placing them in separate culture vessels (containing culture medium with plant growth regulators);;**
4. **The cells of a callus can be induced to (either proliferate or) differentiate into particular tissues e.g. shoot or root by varying the ratio of the various plant growth regulators to form plantlets;;**
5. **(The ratio of plant growth regulators will result in the plant tissues developing into specialized tissues.)**
6. **(Equal ratio of cytokinin and auxin stimulates mitosis.)**
7. **Auxin and gibberellin stimulate cell differentiation / low level of cytokinin and high level of auxin triggers root growth / high level of cytokinin and low level of auxin triggers shoot growth.**

(b) State two environmental conditions that need to be controlled during the development of the plantlets. [1]

1. **light intensity / temperature / humidity / nutrients / pH / water content (any two);;**

(c) Suggest two advantages of propagating banana plants using this system. [2]

1. **Rapid multiplication of banana plants - shoots produced during cloning produce buds which can be used to generate more shoots (using the same tissue culture technique) / banana plants with desired traits can be multiplied rapidly than can be done using conventional breeding methods/via sexual reproduction;;**
2. **Genetic uniformity - banana plants produced via cloning are genetically identical and all possess the desirable features of the stock plants;;**
3. **Production of disease-free plant material - viruses do not normally penetrate the tips of meristems, it is therefore possible to use meristems for producing virus-free banana plants by tissue culture / Since the banana plants are prepared in sterile conditions they are free from surface bacteria and fungi, some of which would cause disease;;**
4. **Effective means of asexual reproduction in banana plant species which are difficult to propagate by germination from seeds / have low seed germination rates;;**
5. **Independent of climate changes - micropropagation is not limited by seasonal changes in climate so banana plants can be produced continuously, giving great flexibility in meeting consumer demand (because banana plants can be produced out of season) / consumers can buy plants at lower prices than usual.**
6. **Ability to air freight large quantities - the large numbers of banana plantlets produced are light and small in size (and can be stored in culture flask) to be air-freighted and transported easily and cheaply, thereby increasing international trade;;**
7. **Sales advantage - batches of identical banana plants can be obtained, ensuring product uniformity which is an important sales advantage;;**
8. **Take up little space - tissue culture takes up relatively little space and can be grown intensively;;**

The effect of plant growth regulators, IAA and Kinetin, on shoot regeneration is shown in Table 3.1.

Table 3.1

MS medium + growth regulators ($mg\,l^{-1}$)		Explants differentiated	Mean number of shoots per explant
+ IAA	+Kinetin		
0	0	29	2
0·1	1·0	42	4
0·1	2·5	46	8
0·1	5·0	58	8
0·5	1·0	54	12
0·5	2·5	66	20
0·5	5·0	58	28

(d) With reference to Table. 3.1,

(i) state the evidence that growth regulators benefit propagation, [1]

1. As concentration of growth regulators increase, the mean number of shoots per explant increase / number of explants differentiated increase;;

(ii) explain why a growth regulator combination of 0·5 mg l^{-1} IAA + 5·0 mg l^{-1} kinetin is recommended. [1]

1. At 0·5 IAA + 5·0 kinetin ➔ 58 × 28 = 1624 shoots = highest yield;;

For explanation: At 0·5 IAA + 2.5 kinetin ➔ 66 × 20 = 1320 shoots

Fig. 3.2 shows the proportion of cotton, maize and soybeans grown in the USA that are genetically modified in two different ways:

- HT crops are modified to be resistant to the glyphosate herbicide.
- Bt crops are modified to express the Bt toxin which kills insect pests.

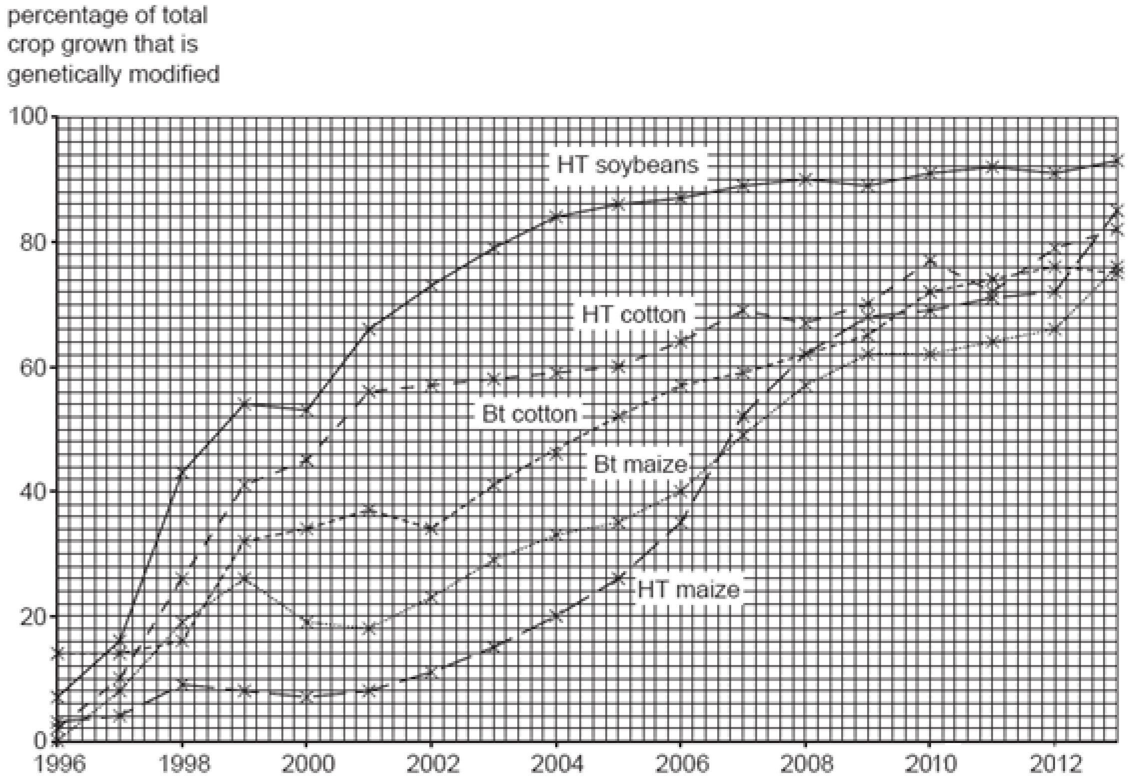


Fig. 3.2

(e) (i) With reference to Fig. 3.2, describe the change in the percentage of genetically modified soybeans grown in the USA since 1996. [2]

1. **HT soybeans increased from 7% (in 1996) to 93% in 2013;;**
2. **steepest / fastest, increase between 1996 and 1999;;**
3. **remain relatively constant at 89 – 93% between 2007 – 2013;;**

Fig. 3.3 shows the increase in the number of weed species resistant to glyphosate herbicide and triazine herbicide since 1970.

Crops have not been genetically modified to resist triazine herbicide.

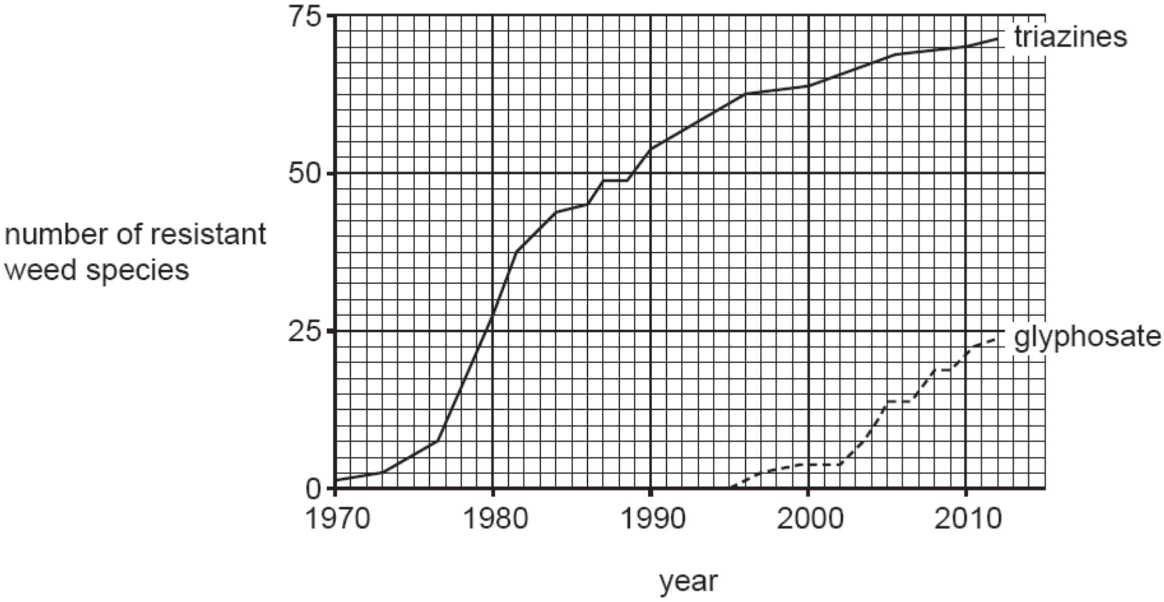


Fig. 3.3

(ii) With reference to Fig. 3.2 and Fig. 3.3, suggest if the rise in glyphosate-resistant weeds has resulted from the introduction of genetically modified crops. [1]

1. **number of glyphosate-resistant weed species only increased after 1995/1996, this was when, GM crops resistant to herbicide/HT crops, were introduced;;**

(iii) Suggest one possible risk to the environment as a result of growing genetically engineered crops. [1]

1. **Genetically engineered crops grow quickly, out-compete wild crops for nutrients, space etc and may establish themselves as weeds. (This can also lead to reduced biodiversity);;**
2. **Unintended transfer of transgenes through cross-pollination from genetically engineered crops to weeds / There is the possibility of the introduced genes escaping from a genetically engineered crops into related weeds through crop-to-weed hybridization;;**
3. **Genetically engineered crops could prove toxic to non-target organisms that may benefit agriculture such as pollinators, leading to reduced numbers or extinction of that organism;;**
4. **Genetically engineered crops may upset the balance of the native species which might eventually upset the ecological balance and lead to decrease in biodiversity;;**

[Total: 12]

4 Planning Question

The green alga, *Scenedesmus quadricauda*, cells contain the photosynthetic pigments chlorophyll a and b. This enables the green algae to undergo photosynthesis.

S. quadricauda cells require carbon dioxide that is dissolved in water to photosynthesize. Water does not naturally contain a lot of dissolved carbon dioxide. When sodium hydrogen carbonate is dissolved in water, it produces carbon dioxide and increases the amount of carbon dioxide in the water. The concentration of hydrogen carbonate ions in a solution is directly proportional to the concentration of carbon dioxide available.

S. quadricauda can be immobilised in alginate beads. The alginate beads are to be arranged in a monolayer to fully cover the bottom of a conical flask.

Using this information and your own knowledge, design an experiment to test the hypothesis that:

"The rate of photosynthesis is dependent on the concentration of carbon dioxide"

You must use:

- an unlimited supply of *S. quadricauda* alginate beads all of diameter 11 mm,
- an unlimited volume of 1.0 % sodium hydrogen carbonate,
- an unlimited volume of distilled water,
- 250 cm^3 conical flasks of diameter 85 mm,
- 100 cm^3 gas syringe.

You may select from the following apparatus:

- any normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, glass rod, etc.,
- syringes,
- 100-watt table lamp,
- timer e.g. stopwatch.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment

[Total: 12]

Mark Scheme

1. Theoretical consideration or rationale of the plan to justify the practical procedure.
2. Identification of variables – independent and dependent variables.
3. Description of method used i.e. measuring oxygen volume collected using a gas syringe
4. 3 identification of other variables to be kept constant.
5. Sodium hydrogen carbonate concentration as measured by at least 5 different concentrations (A! range of 0.0% - 1.0%).
6. Serial dilution table.
7. Description of control.
8. Correct number of beads used (A! 50-59 beads in a monolayer)
9. Equilibrium time.
10. Reading volume of oxygen collected after specific time.
11. Relevant clearly labeled diagram.
12. Replicates and at least two repeats.
13. Calculation of mean/average (average calculated in table or in graph).
14. Layout of results table and graph with appropriate headings and average oxygen collected / cm^3 (y-axis) against concentration of sodium hydrogen carbonate / % (x-axis).
15. 2 safety precautions / risk hazards identified.
16. The correct use of technical and scientific terms.

[12 marks]

Variables [2]

The independent variable in this experiment is concentration of sodium hydrogen carbonate (carbon dioxide) while the dependent variable is volume of oxygen collected (rate of photosynthesis).

Other variables to be kept constant [4]

- Temperature - Use a water bath / glass shield to block heat from lamp.
- Volumes of sodium hydrogen carbonate concentration.
- Amount of algal beads used.
- Equilibrium time of 5 minutes to ensure that oxygen is given off at the beginning of each experiment.
- Light intensity - using same light source at same distance from algal beads.
- Duration of the experiment ??

Prediction of the likely outcome of the experiment [1]

- As concentration of sodium hydrogen carbonate increases, the volume of oxygen collected increase, rate of photosynthesis increase.
- As concentration of sodium hydrogen carbonate increases further, the volume of oxygen collected levels off / remains constant.

Explain the likely outcome of the experiment [1, 16]

- Carbon dioxide (from sodium hydrogen carbonate) is needed in the light independent reactions - used during carboxylation phase for synthesis of carbohydrate.
- An increase in the carbon dioxide concentration increases the rate at which carbon is incorporated into carbohydrate in the light-independent reaction / Calvin cycle.
- Rate of NADPH oxidation / $NADP^+$ regeneration for light reaction increases.
- Light reaction proceeds at a faster rate.
- Rate of photolysis of water increases and more oxygen is evolved per unit time until another factor becomes limiting.

Explanation of how dependent variable can be measured [3]

- Oxygen evolved by plant materials is collected in the gas syringe and volume is read off from the gas syringe after 5 minutes.
- The higher the volume of oxygen collected means the rate of photosynthesis is higher.

Procedure(includes control and reliability of results)

1. Prepare 250 cm^3 each of 0.2%, 0.4%, 0.6%, 0.8%, 1.0% sodium hydrogen carbonate solution and a control of 0.0% sodium hydrogen carbonate by diluting the 1.0% stock sodium hydrogen carbonate solution as indicated in the table below. [5]

Specimen tubes	Concentration of sodium hydrogen carbonate / %	Volume of 1.0% sodium hydrogen carbonate / cm^3	Volume of distilled water / cm^3
A	0.2	50.0	200.0
B	0.4	100.0	150.0
C	0.6	150.0	100.0
D	0.8	200.0	50.0
E	1.0	250.0	0.0
Control [7]	0.0	0.0	250.0

[5, 6]

2. Placed them in labelled conical flasks.

Setting up the experiment

3. Place A! 50 - 59 algal beads in a monolayer in a conical flask filled with 0.2% sodium hydrogen carbonate solution. [8]

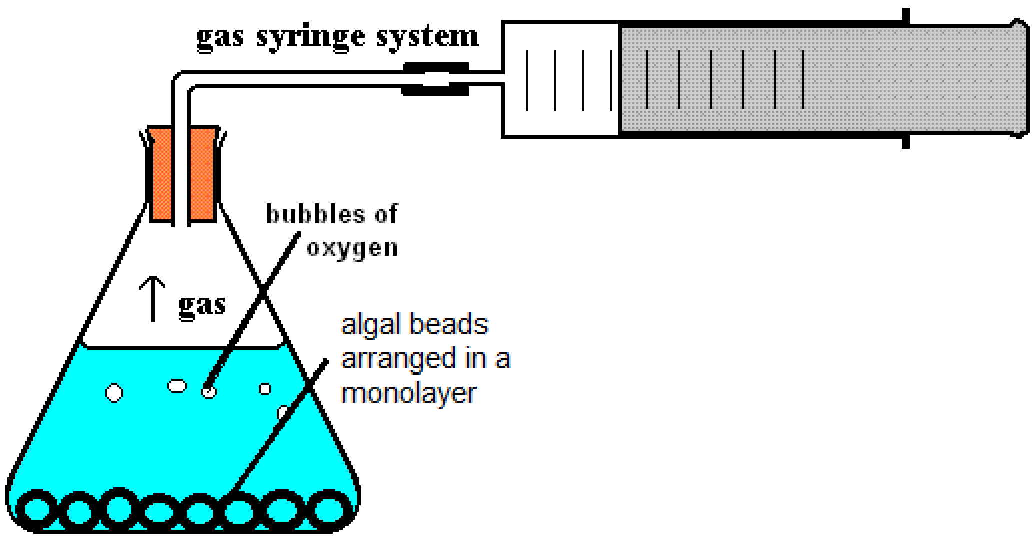


Fig. 1 [11]

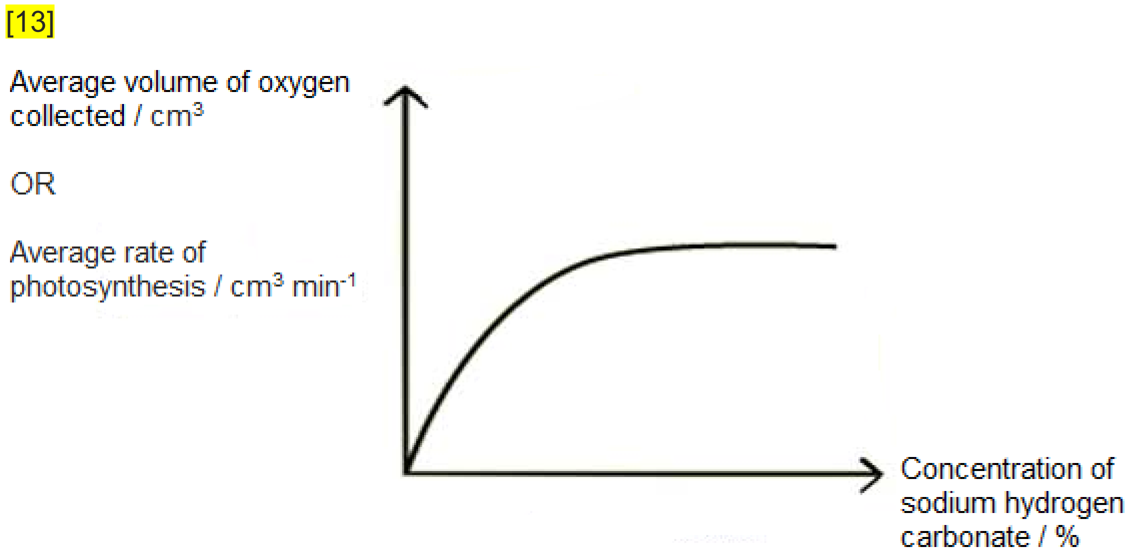
4. Set up the apparatus as shown in Fig.1.
5. Put a 100-watt table lamp at 10 cm from the algal beads. Turn on the lamp.
6. Wait for about 5 minutes for the algal beads to equilibrate / for oxygen to be released. [9]
7. After 5 minutes, record the final reading in the gas syringe and calculate the volume of gas collected. [3, 10]
8. Repeat step 1-7 to obtain another 2 readings. [12]

9. Repeat step 1-8 for 0.4 %, 0.6 %, 0.8%, 1.0%, 0.0 % (control) [12]
10. Repeat entire experiment twice. [12]
11. Plot a graph of volume of oxygen collected / cm^3 against concentration of sodium hydrogen carbonate / %.

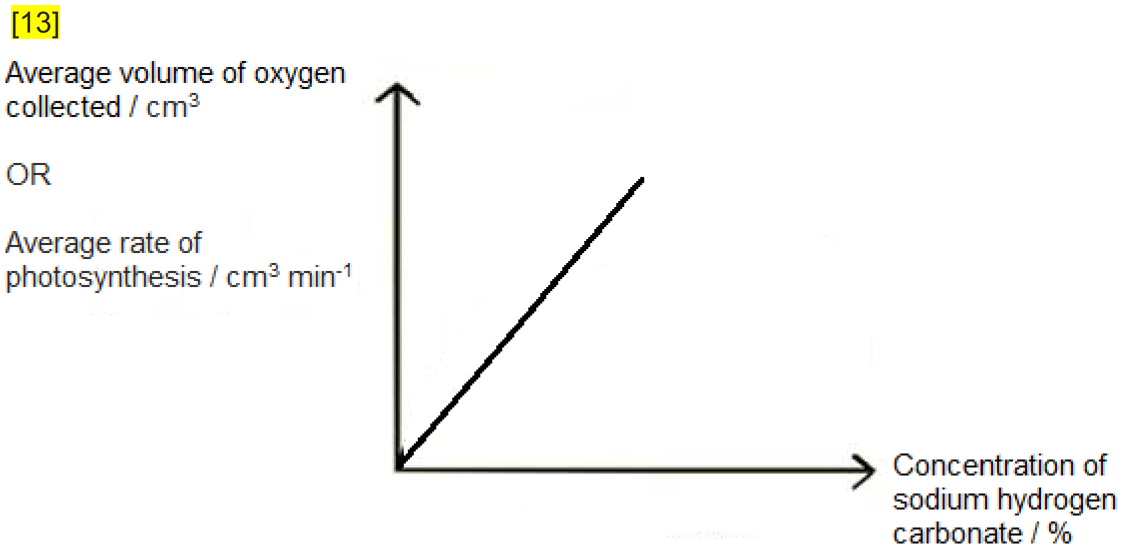
<u>Table showing volume of oxygen collected at varying sodium hydrogen carbonate concentration [14]</u>

Concentration of sodium hydrogen carbonate / %	Volume of oxygen collected / cm^3				Average rate of photosynthesis / cm^3 min^{-1} [13]
	Reading 1	Reading 2	Reading 3	Average [13]	
0.2					
0.4					
0.6					
0.8					
1.0					
0.0 (control)					

<u>Graph showing volume of oxygen collected / cm^3 against concentration of sodium hydrogen carbonate / %</u> [14]



OR



<u>Risk Assessment</u> [15]

Risks	Safety Precautions
Handle liquid chemicals like sodium hydrogen carbonate solution with care.	Wear protective goggles;;
Handle fragile objects / glassware with care.	Arrange conical flask / gas syringe neatly ensuring that they are placed where they cannot be knocked over or rolled off the bench;;

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Describe the polymerase chain reaction (PCR) and explain how PCR facilitates genetic fingerprinting. [8]

1. Denaturation by heating to 95°C;;
2. hydrogen bonds between double stranded DNA break, separating the double-stranded DNA into single stranded DNA;;
3. Annealing of DNA primers by cooling to 65°C;; (A) 30-65°C
4. Primers base pair via complementary base pairing with complementary sequences at the 3' end of single stranded DNA;;
5. primer extension by heating to 72°C;; (A) 60-75°C
6. Taq polymerase adds free nucleotides / dNTPs to 3' ends on both primers to synthesize rest of the fragment (new strands of DNA);;

7. PCR generate sufficient DNA for the various procedures involved in genetic fingerprinting;;
AND
8. High sensitivity: PCR can amplify sequences from minute amounts of DNA, this is especially useful in forensic work where DNA may not be present in high quality;;
9. Speed: Millions of copies of target DNA can be obtained in a few hours;;
10. Robustness: PCR can permit amplification of specific sequences from material in which the DNA is badly degraded or embedded in a medium;;
11. Automation: use of Taq polymerase which can withstand high temperatures without being denatured (thermal stable) allows for the automation of PCR;;
(pt 8 – 11, max 3)

(b) Explain how RFLP analysis facilitated the process of DNA fingerprinting. [6]

1. **DNA fingerprinting is a technology that identifies particular individuals using properties of their DNA;;**
2. **Non-coding sequence at a particular locus on a chromosome;;**
3. **may exhibit variable-number tandem repeats (minisatellites);;**
4. **The number of such tandem repeats varies among individuals;;**
5. **and is a form of DNA polymorphism;;**
6. **Digestion of DNA (with the same restriction enzymes) from different individuals will generate restriction fragments;;**
7. **that are of different molecular weight / length / size, giving rise to a unique DNA fingerprint;;**
8. **These fragments can be separated using gel electrophoresis: Rate of migration is inversely proportional to the length/size/molecular weight of DNA fragment;;**
9. **Southern blot can then be carried out using a probe complementary to part of the VNTR region;;**
10. **DNA fingerprint of an individual can be compared against another individual to detect matching band pattern;;**
11. **for forensic analysis for crime investigation / to determine paternity / trace evolutionary lineage;;**

(c) Discuss the ethical concerns that have arisen about the human genome project. [6]

1. **<u>Privacy and confidentiality of personal genetic information:</u> It is difficult to determine who owns the genetic information—the individual who donated the sample, the hospital, or the company which sequenced the information. There is also an issue who should have access to the personal genetic information; or if the information should be kept confidential to be accessed by the individual only;;**
2. **<u>Fairness in the use of personal genetic information</u>: It is of concern that the information may be used in a discriminatory manner. For example, employers/insurance companies have access to the personal genetic information and hire workers/grant insurance policies based on their genetic predisposition;;**
3. **<u>Psychological impact and stigmatization due to genetic differences</u>: It is unclear whether the personal genetic information that is revealed may affect society's perceptions of that individual or how the information may affect members of minority communities;;**
4. **<u>Reproductive issues regarding use of genetic information in reproductive decision making and reproductive rights</u>: There are concerns that prenatal genetic testing could lead to genetic manipulation or a decision to terminate pregnancy based on undesirable traits or presence of genetic disorders disclosed by the genetic testings;;**
5. **<u>Clinical issues</u>: There are questions on the accuracy, reliability and utility of the genetic tests and how much the interpretation can be trusted. False-positive or false-negative results can occur due to technical abnormalities or human error. There is also an issue of whether healthcare personnel are properly counselling people about the risks and limitations of genetic technology;;**
6. **<u>Fairness in access to advanced genomic technologies</u>: It is difficult to ascertain who will benefit or if it will result in major worldwide inequities. For example, the designer drugs developed by pharmacogenomics may discriminate against the poor as the drugs may be expensive and only the wealthy can afford them;;**
7. **<u>Commercialisation of products including property rights and accessibility of data and materials</u>: It is difficult to determine who owns genes and other pieces of DNA and if they can be patented by companies/individuals. Patents could limit the accessibility of the data because of the cost associated with using patented research data. This could limit the development of useful products from the gene information;;**
8. **<u>Conceptual and philosophical implications regarding human responsibility, free will vs genetic determinism, and concepts of health and disease</u>: Knowledge of whether genes alone affect a person's behaviour are not well studied and it is hard to make a clear distinction between medical treatment and enhancement, as well as what is considered a disorder and what is within the limits of normality;;**
9. **<u>Uncertainties associated with gene tests</u>: Uncertainties include the issue of whether testing should be performed when treatment is unavailable or too expensive and unaffordable as this may create anxiety and frustration. It is also unclear whether parents have the right to have their children tested for adult-onset diseases as there could be conflict between parent's choice and a child's welfare. Another issue would be determining whether people who are incapable of giving valid consent, e.g. newborns and the mentally incapacitated, should undergo genetic screening;;**

[Total: 20]